BELL 206 JET RANGER COCKPIT

Jet Ranger Quick Reference

One-page summary of the cockpit wiring, network and controller boards. Built from the project documentation as it stood on 8 November 2026 (firmware 1.0). The full detail is in the master document and the Cockpit User Guide.

01 Cockpit at a glance

A home-built Bell 206 Jet Ranger cockpit. Real gauges, warning lamps and switch panels are driven from a flight simulator by Arduino Mega boards on a private Ethernet network.

SIMULATORS
MSFS 2020/2024 (SimConnect), Prepar3D / FSX (FSUIPC), DCS World (DCS-BIOS), X-Plane (UDP datarefs)

DATA PATH
Simulator, bridge program on the PC, UDP text packets about 10 times a second, Arduino boards

CONTROLLERS
5 Mega 2560 boards plus 1 Leonardo USB HID board, all with W5500 Ethernet

NETWORK
Wired `172.16.1.0/24`, fixed addresses, no DHCP. PC must be `.10`

RUN ONE BRIDGE
FSUIPC and MSFS both use UDP 27001; all four send to the same boards

PLATFORM
CKAS 6DOF motion platform, USB to the PC

CONTROLLER	IP	WHAT IT RUNS
Radio	172.16.1.101	COM and NAV displays, eight tuning knobs, four swap switches, Auto Pilot and Audio panels, DME display, Huey Warning Panel
Servo	172.16.1.102	Attitude indicator pitch and bank servos, 15 warning lamps, TOT clear, warning mute and fuel valve switches
Stepper	172.16.1.105	Stepper gauges: airspeed, VSI, radar altitude, torque, gas producer, turbine out temperature, fuel quantity, oil gauges
Dual Stepper	172.16.1.106	Engine and rotor RPM, electrical and fuel load, altimeter; four OLEDs (barometer, altimeter, clock, temperature)
ADF and Upper	172.16.1.107	ADF knobs and keypad, 7 lower-panel switches, landing gear lamps, overhead switch matrix (72 positions), Upper keypad
USB HID board	172.16.1.11	Turns UDP button and key commands into four 32-button joysticks and a keyboard

02 IP address and port mapping

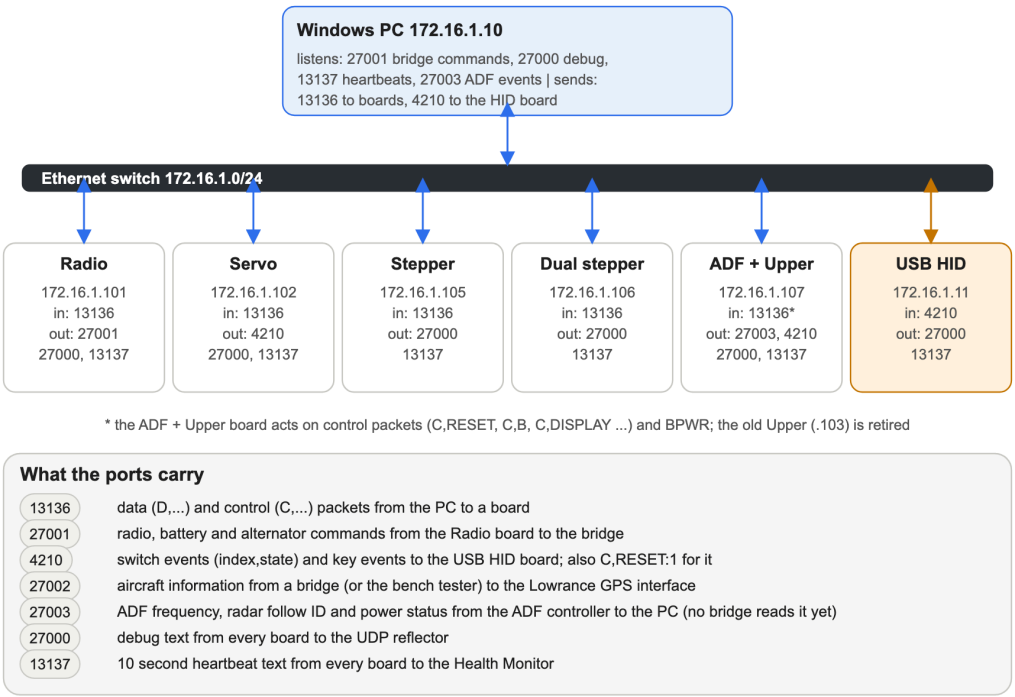
PORT	DIRECTION	PURPOSE
13136	PC to boards	Data (`D,CODE:value,...`) and control (`C,COMMAND:value`) packets
13137	Boards to Health Monitor	Heartbeat text every 10 s, with 5 V rail and firmware version
13138	Stepper to other controllers	`STARTUPDONE`, sent twice 2 s apart; ends the start-up backlight hold (180 s fallback)
27000	Boards to PC	Debug text, each line prefixed with the board name
27001	Radio and ADF controllers to bridge	Standby frequencies, radio swaps, Auto Pilot, Audio and ADF panel commands. FSUIPC and MSFS bridges only
27002	Bridge to GPS interface	Position, altitude, airspeed, heading and simulator UTC every 500 ms
27003	ADF controller to PC	`ADF_FREQ`, radar follow ID, power status every 30 s, `UPPER_KEY:`, `SQUAWK_SET:`

PORT	DIRECTION	PURPOSE
4210	Servo, Radio, ADF and Upper to HID board	Button changes (`index,state`), keys, reset
5010	DCS to DCS bridge	DCS-BIOS export stream, multicast 239.255.50.10
49000	X-Plane and X-Plane bridge	Dataref requests and replies; beacon is multicast 239.255.1.1 port 49707
13136 on 127.0.0.101 to .107	Bridge to panel emulators	Same board packets, sent to the emulators on the PC
7788	Local on the boards	Source port for outgoing packets

PACKET CHEAT SHEET

- Gauge codes take real units (degrees, knots, feet per minute, psi, percent, degrees C, US gallons). Add `RAW` to a code to skip the calibration table (`IASRAW:300`).
- Lamps take 0 or 1. Example: `D,PITCH:5,BANK:-10,LOWF:0` sent to 172.16.1.102 port 13136.
- Common controls: `C,B:0-15` brightness, `C,RESET:1` reboot, `C,VERSION:1` report firmware, `BPWR:0/1` backlight power.

03 Network



- All devices are on one wired `172.16.1.0/24` network with fixed addresses. No DHCP.
- Boards send debug and heartbeat traffic to `172.16.1.10`. The primary PC must hold that address, or `reflectorIP` must be changed in every sketch.
- Only one program can own a UDP port. The Health Monitor owns 13137, each bridge owns 27001, and the debug reflector owns 27000.
- The network switch sits in the front panel assembly and is fed by its own 5 V cable from the ATX supply PCB.

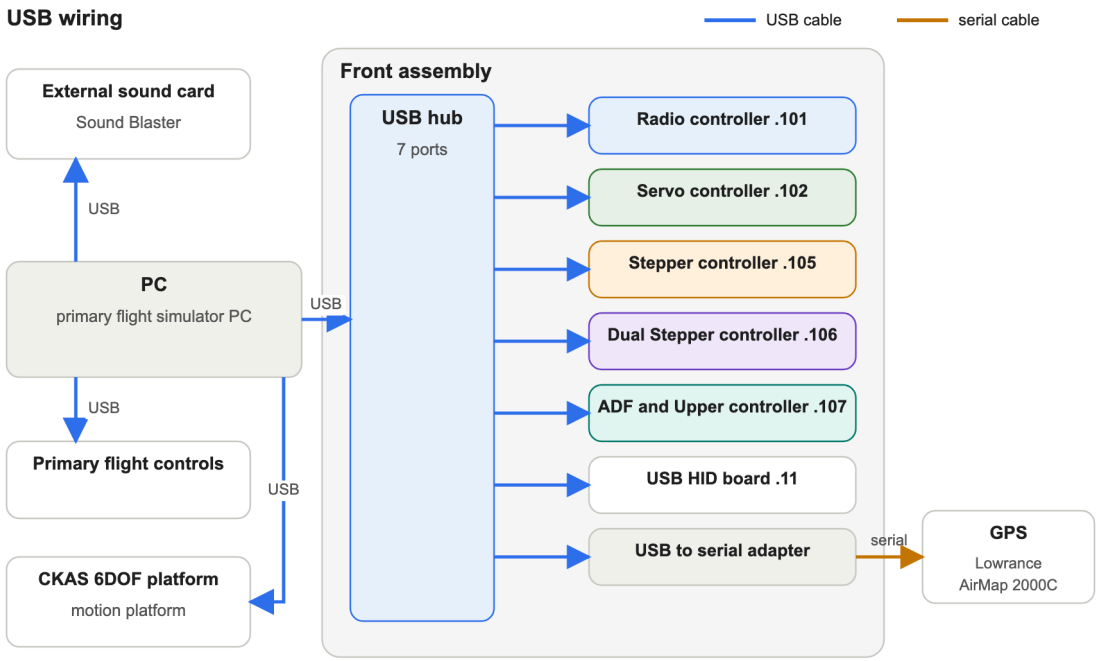
> **Bursty data with gaps of several seconds** means the switch is dropping its ports. On this cockpit the cause was the 5 V rail feeding the switch dipping below 4.75 V. Check that supply first, then the cable and the switch port.

BOARD BEHAVIOUR TO KNOW

- **No data for 30 s:** gauges return to zero, Servo lamps go off, Dual Stepper OLEDs and Radio displays blank, backlight goes off.

- **Power rules in the bridges:** main bus below 20 V sends `BPWR:0` (backlights off, Servo lamps off, radar altitude zero). Main bus below 24 V blanks the Dual Stepper OLEDs. Avionics bus below 24 V blanks the Radio COM and NAV displays.
- **Start-up hold:** every controller holds its backlight at 100 percent, with OLEDs and lamps on, until the Stepper announces `STARTUPDONE`. Backlight then goes to 70 percent once `BPWR:1` arrives.
- **Remote reset:** `C,RESET:1` reboots a board after about 2 s. The HID board briefly drops all four joysticks and the keyboard.

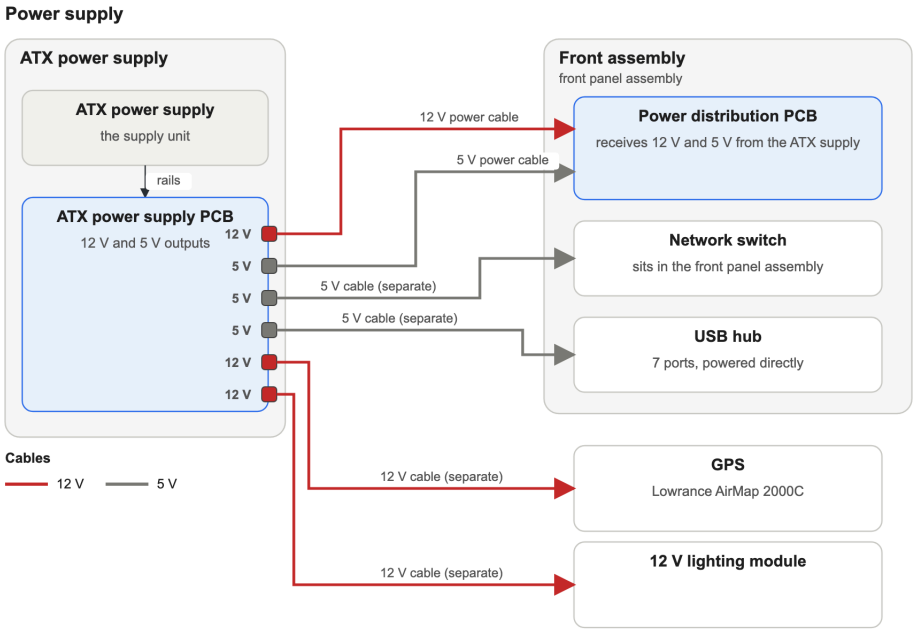
04 USB wiring



The sound card, the flight controls and the CKAS platform connect to the PC directly, not through the hub.

- A **7 port USB hub** in the front assembly connects to the PC. It is powered directly from the ATX supply PCB, not through the power distribution PCB.
- On the hub: the Radio, Servo, Stepper, Dual Stepper, and ADF and Upper controllers, the USB HID board, and a USB to serial adapter that runs to the GPS.
- Straight to the PC on their own USB cables: the **Sound Blaster**, the **primary flight controls** and the **CKAS 6DOF platform**.

05 Power supply



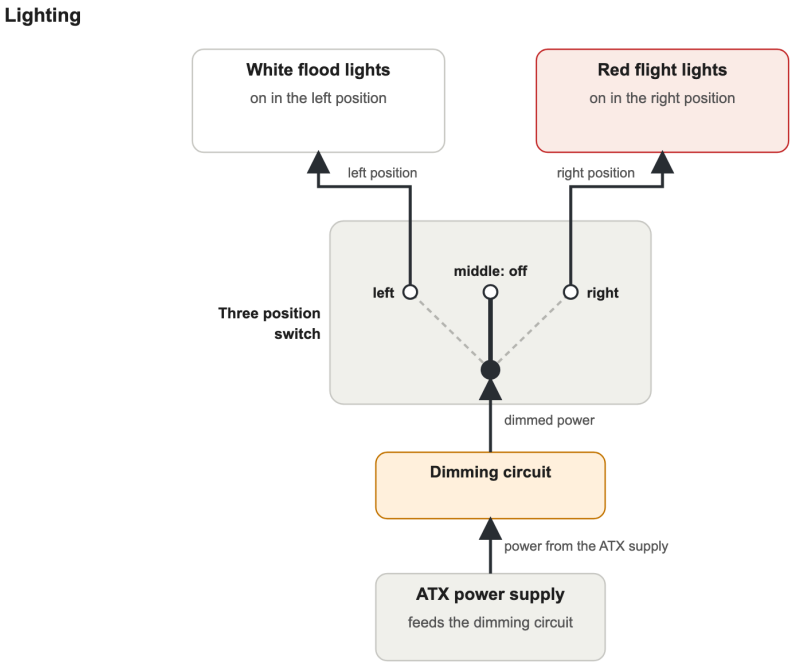
The ATX supply PCB feeds the power distribution PCB with its own 12 V and 5 V cables.
The GPS, the 12 V lighting module, the network switch and the USB hub each have a separate cable of their own.
The USB hub is powered directly, not through the power distribution PCB. Each cable is one line; ground returns are not drawn.

FROM THE ATX SUPPLY PCB	GOES TO
12 V and 5 V (two cables)	Power distribution PCB in the front assembly
12 V	GPS (separate cable)
5 V	Network switch in the front panel assembly (separate cable)
5 V	USB hub in the front assembly (separate cable)
12 V	12 V lighting module (separate cable)

Each controller board takes a 5 V input (J2/J4 DC jack) and a 12 to 20 V input labelled STEPPER IN (J3/J7 DC jack). The 12 V input feeds lamps and backlights.

> **Open:** the Health Monitor showed every board's 5 V rail between 3.9 and 4.3 V on 2026-11-01, below the 4.75 V limit. Measure at a board's 5 V pin to see whether it is the supply or the reading (the reading uses the ADC reference). The ATX supply PCB is not yet installed.

06 Lighting



The switch is drawn in the middle (off) position. The supply voltage, the dimming circuit's design and the wiring of the lights are not drawn.

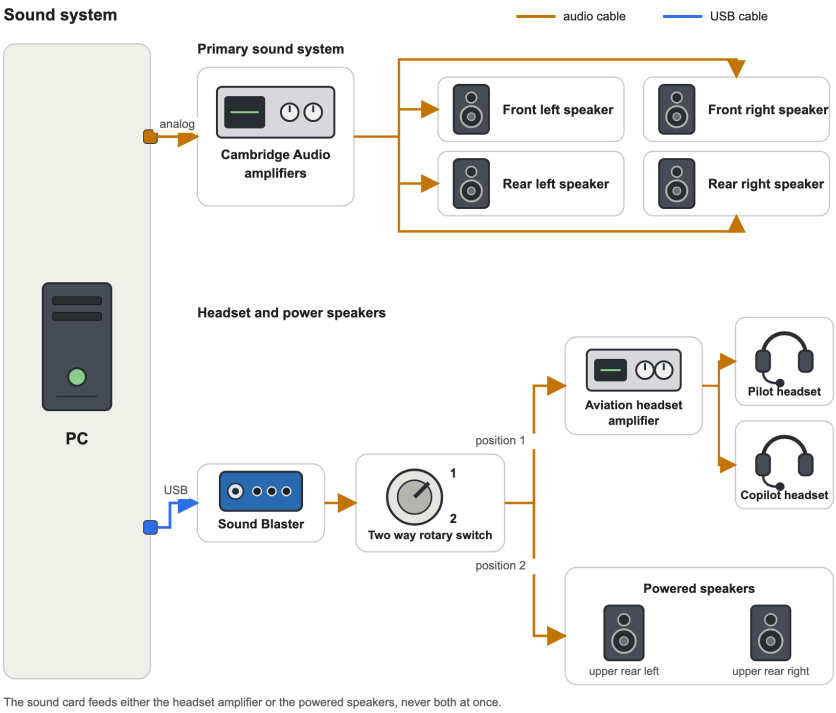
THREE POSITION SWITCH	RESULT
Left	White flood lights on
Middle	Off
Right	Red flight lights on

The ATX supply feeds a dimming circuit, and the dimming circuit feeds the three position switch, so it dims whichever lights are on. The ATX supply PCB also gives the 12 V lighting module its own cable.

PANEL AND GAUGE BACKLIGHTS

- Every controller drives its backlight from **Mega pin D5** (PWM) through a 330 ohm resistor to an **IRF540N** MOSFET, low side. `C,B:0–15` sets the level; `BPWR` switches it with the main bus.
- The brightness pot on the ADF and Upper board (A0, connector J16) is wired but switched off in the sketch (`USE_BRIGHTNESS_POT` 0).

07 Sound

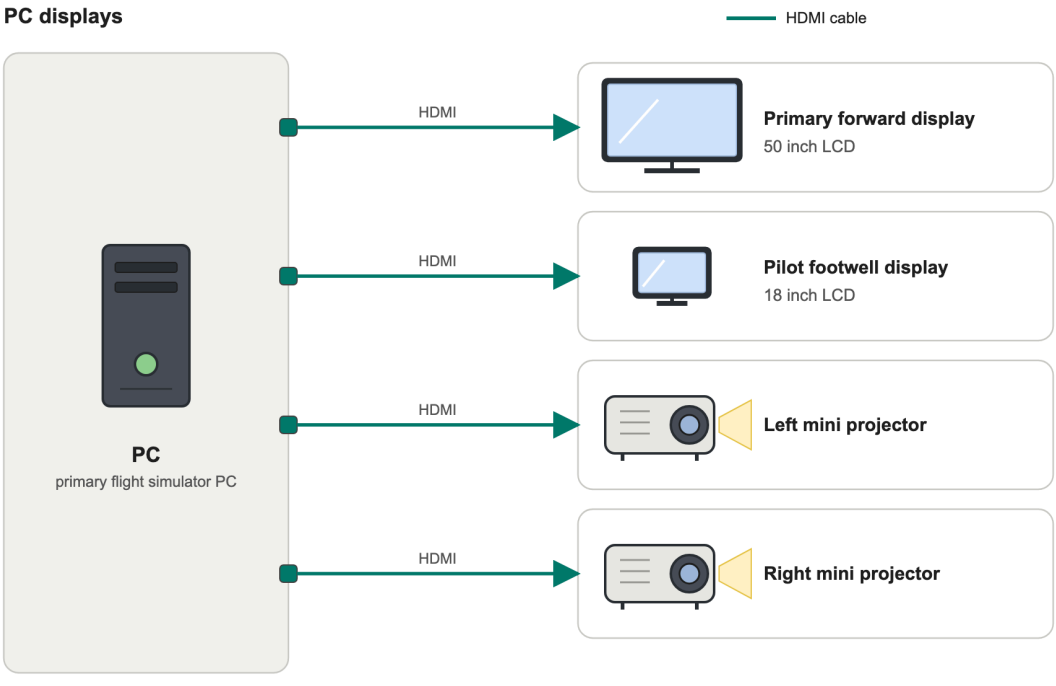


The sound card feeds either the headset amplifier or the powered speakers, never both at once.

SYSTEM	PATH
Primary	PC analog outputs to Cambridge Audio amplifiers to four speakers: front left, front right, rear left, rear right
Headset and powered speakers	USB Sound Blaster to a two way rotary switch. Position 1: aviation headset amplifier (pilot and copilot headsets). Position 2: powered speakers at the upper rear left and right of the pit

The Sound Blaster feeds one of the two at a time.

08 PC displays



All four displays are connected to the PC by HDMI.

DISPLAY	SIZE / TYPE	CONNECTION
Primary forward display	50 inch LCD	HDMI
Pilot footwell display	18 inch LCD	HDMI
Left mini projector	Projector	HDMI
Right mini projector	Projector	HDMI

09 IP address, MAC and COM ports

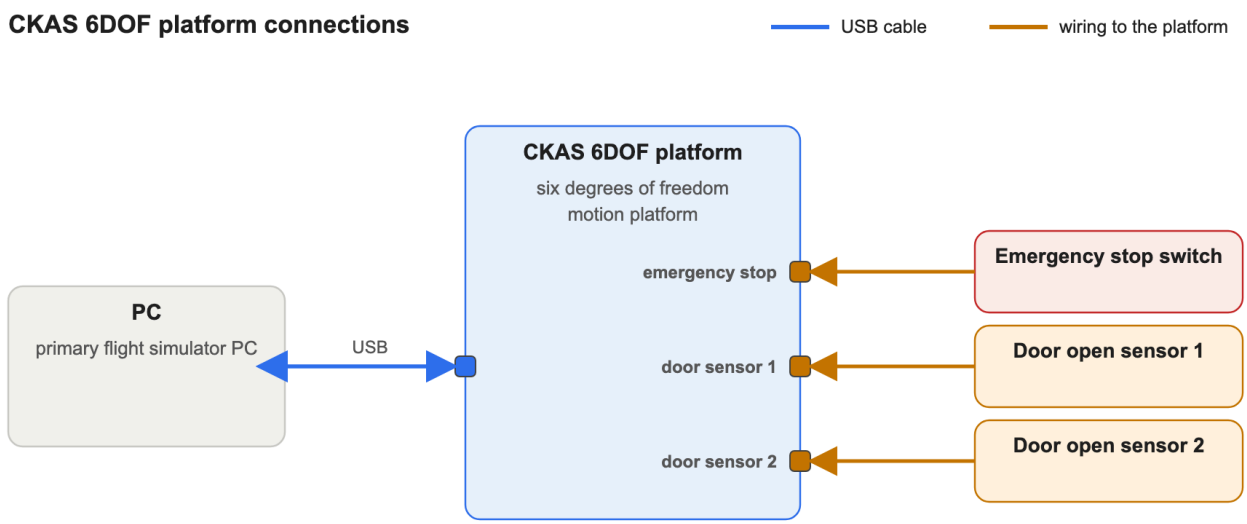
DEVICE	IP	MAC	HEARTBEAT	HARDWARE	COM
Primary PC	172.16.1.10				
USB HID board	172.16.1.11	DE:AD:BE:EF:FE:ED	JOYSTICK	Arduino Leonardo	COM8
Radio controller	172.16.1.101	A8:61:0A:9E:83:01	COMM_NAV	Mega 2560 (replacement)	COM20
Servo controller	172.16.1.102	A8:61:0A:9E:83:02	SERVO	Mega 2560 clone	COM16
Stepper controller	172.16.1.105	A8:61:0A:67:83:69	STEPPER	Mega 2560	COM4
Dual Stepper controller	172.16.1.106	A8:61:0A:67:83:6A	DUAL_STEPPER	Mega 2560	COM5
ADF and Upper controller	172.16.1.107	A8:61:0A:67:83:6B	ADF_PANEL, UPPER_INPUT	Mega 2560	COM26

- **Retired:** Upper controller 172.16.1.103 (MAC A8:61:0A:9E:83:03) merged into the ADF controller on 31 October 2026. The bridges and BenchTester still send to it; those entries are on the to-do list.
- **COM numbers** are those used at the last flash. They follow the USB socket and the board. The Servo clone has no USB serial number, so its COM number is tied to the socket. A replaced Mega gets a new number. Check before every upload.
- The GPS serial port is asked for on first run of the GPS interface and remembered (COM25 on one PC, which was absent on another).
- The BenchTester *COM ports...* window lists ports with board type and serial number.

FIRMWARE NOW (VERSION 1.0)

CONTROLLER	SKETCH	BUILD
Radio	JET_RANGER_RADIO_CONTROLLER	15
Servo	JET_RANGER_SERVO_CONTROLLER	6
Stepper	JET_RANGER_STEPPER_CONTROLLER	5
Dual Stepper	JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER	7
ADF and Upper	JET_RANGER_ADF_CONTROLLER	13
USB HID	UDP_TO_HID	2

10 CKAS 6DOF wiring



The emergency stop switch and the two door open sensors connect directly to the platform.
Not drawn: the platform's own power, the cable types, and what the platform does when an input trips.

- Connected to the PC by its **own USB cable**, not through the hub.
- Wired to the platform: an **emergency stop switch** and two **door open sensors** (door sensor 1 and door sensor 2).

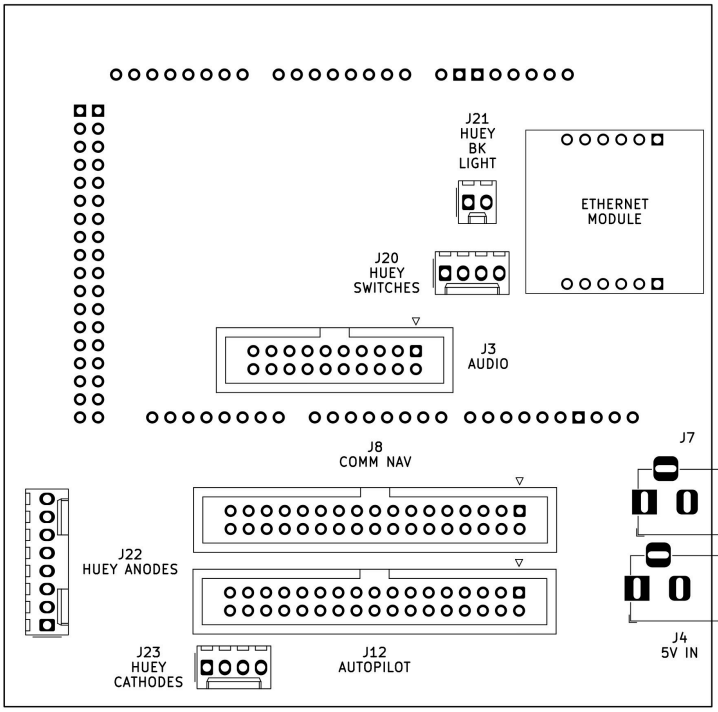
The documentation does not record the platform's own power, the cable types, or what the platform does when one of these inputs trips. Add them here when known.

11 Controller PCB connections

Pin 1 of each connector is the square pad. Backlight MOSFET drive is on D5 of every controller. Green and red status LEDs are on D14 and D15 on every board. The W5500 reset is on D53.

RADIO AUTO PILOT 1.4

Radio controller, 172.16.1.101



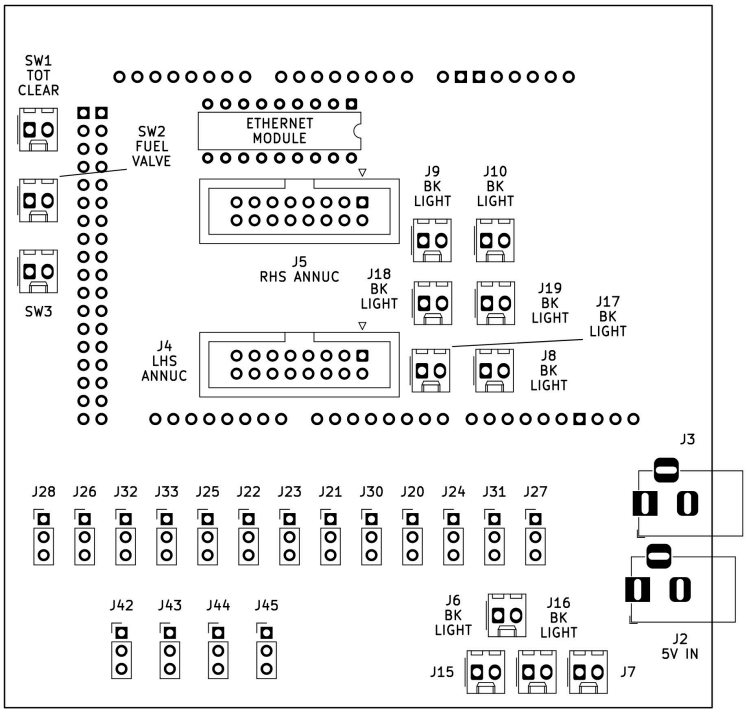
REF	NAME	TYPE	NOTES
J3	AUDIO	IDC 20	Audio panel
J4	5V IN	DC jack	5 V supply
J7	STEPPER IN 12-20VDC	DC jack	12 to 20 V supply
J8	COMM NAV	IDC 34	COM and NAV panel
J12	AUTOPILOT	IDC 34	Auto Pilot panel
J20	HUEY SWITCHES	KK-254 4	GND, dim (A8), test (A9), reset (A10)
J21	HUEY BK LIGHT	KK-254 2	+12 V and switched backlight
J22	HUEY ANODES	KK-254 8	LED anodes 00 to 07
J23	HUEY CATHODES	KK-254 4	LED cathodes 04 to 07

MEGA PIN	FUNCTION
22-25	Matrix rows ROW0 to ROW3
26-29, 48	Matrix columns COL0 to COL4
31,30 / 33,32	COM1 major / minor encoder (A,B)
34,35 / 37,36	COM2 major / minor
38,39 / 41,40	NAV1 major / minor
42,43 / 45,44	NAV2 major / minor
11,12	Auto Pilot CRS knob
A4,A5	ALT knob
46,47	VS knob
7,6	IAS knob
8,9	HDG knob

MEGA PIN	FUNCTION
A15, A14, A13	MAX7219 DIN, CLK, LOAD (panel LEDs)
SDA, SCL	TCA9548A mux 0x70: NAV OLED port 0, COM OLED port 1

FRONT PANEL SERVO 1.4

Servo controller, 172.16.1.102

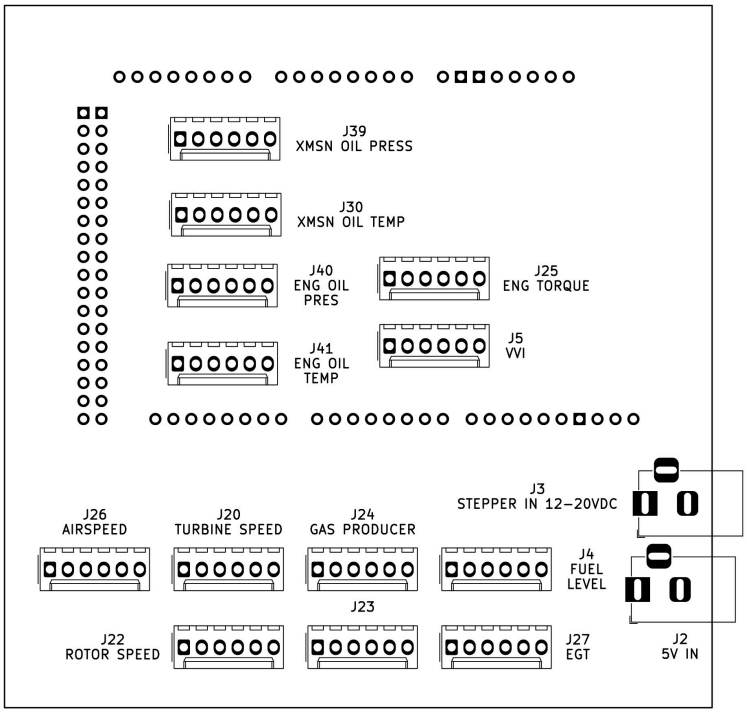


REF	NAME	TYPE	NOTES
J2	5V IN	DC jack	5 V supply
J3	STEPPER IN 12-20VDC	DC jack	12 to 20 V supply
J4	LHS ANNUC	IDC 16	Left annunciator panel
J5	RHS ANNUC	IDC 16	Right annunciator panel
J6-J10, J15-J19	BK LIGHT	KK-254 2	Ten backlight connectors
J20-J33, J44, J45	Gauge headers	Pin header 3	Airspeed, EGT, turbine speed, radar alt, torque, rotor speed, VVI, oil, fuel, gas producer, xmsn
J42 / J43	PITCH / ROLL	Pin header 3	Attitude indicator servos (pins 26, 27)
SW1	TOT CLEAR	KK-254 2	D43, to ground when pressed
SW2	FUEL VALVE	KK-254 2	D48, to ground when on
SW3	WARNING MUTE	KK-254 2	D47, to ground when pressed

MEGA PIN	FUNCTION
A1-A15	Warning lamps (several differ from the sketch order; test each lamp singly)
26, 27	Pitch and bank servos

13 STEPPER 1.6

Stepper controller, 172.16.1.105



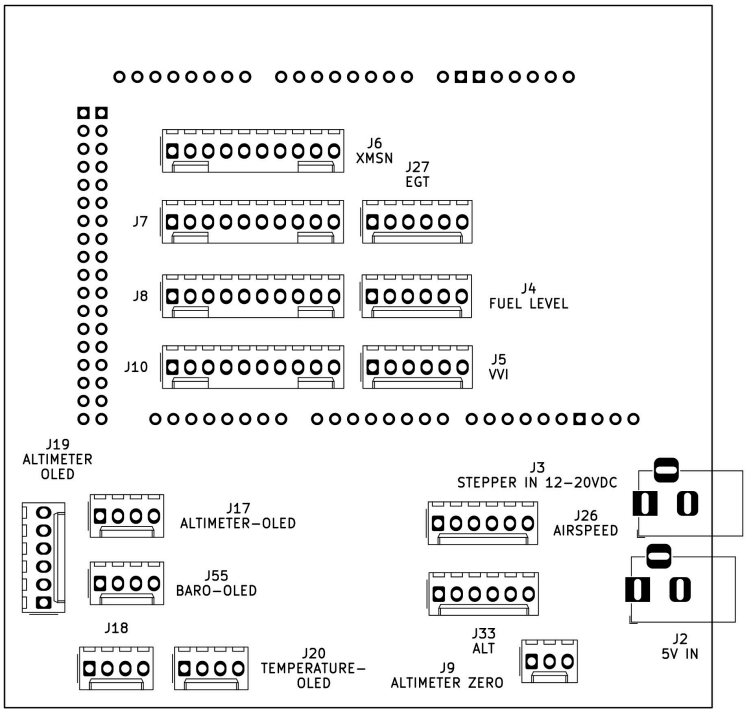
REF	NAME	TYPE	NOTES
J2	5V IN	DC jack	5 V supply
J3	STEPPER IN 12-20VDC	DC jack	12 to 20 V supply
J4	FUEL LEVEL	KK-254 6	
J5	VVI	KK-254 6	
J20	TURBINE SPEED	KK-254 6	
J22	ROTOR SPEED	KK-254 6	
J23	RADAR ALT	KK-254 6	
J24	GAS PRODUCER	KK-254 6	
J25	ENG TORQUE	KK-254 6	
J26	AIRSPEED	KK-254 6	
J27	EGT	KK-254 6	
J30	XMSN OIL TEMP	KK-254 6	
J39	XMSN OIL PRESS	KK-254 6	
J40	ENG OIL PRES	KK-254 6	
J41	ENG OIL TEMP	KK-254 6	

MEGA PIN	FUNCTION
7, 8, 9, 11	VSI coils
12, 13, 22, 23	Airspeed
32-35	Radar altitude
36-39	Torque
40-43	Gas producer

MEGA PIN	FUNCTION
A11-A14	Turbine out temperature
2, 3, 4, 6	Fuel quantity
48, A0-A2	Engine oil temperature
A3-A6	Transmission oil temperature
A7-A10	Transmission oil pressure
44-47	Engine oil pressure

OLED DUAL STEPPER 1.5

Dual Stepper controller, 172.16.1.106



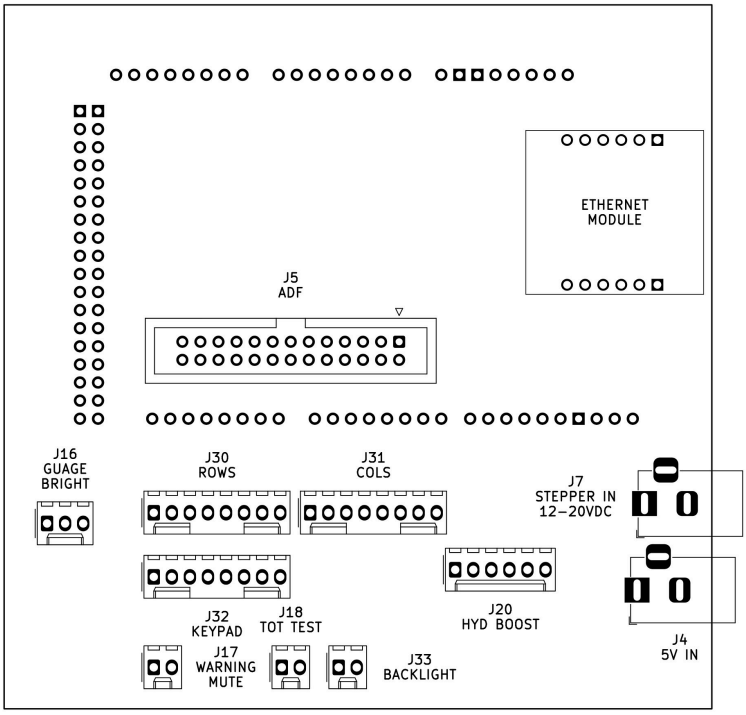
REF	NAME	TYPE	NOTES
J2	5V IN	DC jack	5 V supply
J3	STEPPER IN 12-20VDC	DC jack	12 to 20 V supply
J4	FUEL LEVEL	KK-254 6	
J5	VVI	KK-254 6	
J6	XMSN	KK-254 10	
J7	ROTOR ENG SPEED	KK-254 10	
J8	ENGINE	KK-254 10	
J9	ALTIMETER ZERO	KK-254 3	
J10	ELEC FUEL LOAD	KK-254 10	
J17	ALTIMETER-OLED	KK-254 4	OLED
J18	CLOCK-OLED	KK-254 4	OLED
J20	TEMPERATURE-OLED	KK-254 4	OLED
J55	BARO-OLED	KK-254 4	OLED
J19	ALTIMETER OLED	KK-254 6	
J26	AIRSPEED	KK-254 6	
J27	EGT	KK-254 6	
J33	ALT	KK-254 6	

MEGA PIN	FUNCTION
16	TCA9548A reset
20, 21	I2C SDA, SCL
24-27	Engine speed coils

MEGA PIN	FUNCTION
28-31	Rotor speed coils
32-35	Fuel load coils
36-39	Electrical load coils
40-43, A15	Altimeter coils and zero sensor
44-47, 48, A0-A2	Oil pressure and temperature
A3-A10	Transmission temperature and pressure
2, 3, 4, 6, A11-A14	Fuel quantity and turbine out temperature

UPPER ADF CONTROLLER 1.0

ADF and Upper controller, 172.16.1.107 (combined board, replaces the old Upper and ADF boards)



REF	NAME	TYPE	NOTES
J4	5V IN	DC jack	5 V supply
J7	STEPPER IN 12-20VDC	DC jack	12 to 20 V supply
J5	ADF	IDC 26	ADF panel: knobs, keypad, switches, lamps
J16	GUAGE BRIGHT	KK-254 3	Brightness pot (A0), switched off
J17	WARNING MUTE	KK-254 2	Matrix ROW8 (index 88)
J18	TOT TEST	KK-254 2	Matrix ROW8 (index 89)
J20	HYD BOOST	KK-254 6	Hyd boost D35, anti-ice D36, caution dim D37
J30	ROWS	KK-254 8	Switch matrix rows, D22 to D30
J31	COLS	KK-254 8	Switch matrix columns, D38 to D45, 10 k pull-ups
J32	KEYPAD	KK-254 8	Upper keypad, A9 to A15
J33	BACKLIGHT	KK-254 2	Panel backlight (D5)

MEGA PIN	FUNCTION
D2, D3, D4	Landing gear lamps: nose, left, right
D6, D7 / D8, D9	ADF major / minor encoder (A,B)
D20, D21	I2C: character OLED at 0x3C
D31 to D34	Gear, flaps, speed brakes, brakes switches
A1 to A7	ADF keypad (J41)
A8	12 V monitor (47 k / 10 k divider)
D49	Reset header (J2)

RULES THAT BITE

- Servo PCB lamp pins differ from the sketch for 10 of the 15 lamps (A1/A2, A5/A6, A11/A12 swapped, A7 to A10 reordered). Test lamps one at a time.
- Dual Stepper boards older than 1.4 have D2 and D32 joined: fit only one of fuel quantity and fuel load, or cut the trace.
- Radio encoders COM1 major, COM1 minor, COM2 minor and NAV1 minor have A and B reversed against the sketch.
- The Dual Stepper's RAM use is 79 percent. Be careful adding code.

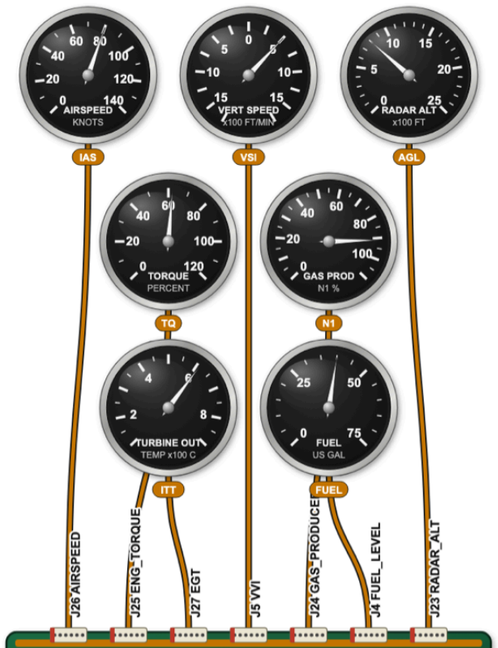
12 Instrument map

Every gauge, lamp and display, and the connector on the controller board that drives it. The Upper panel, the keypads and the Huey switches are drawn as inputs. The A3 original is `Jet Ranger Instrument Map A3.pdf`.

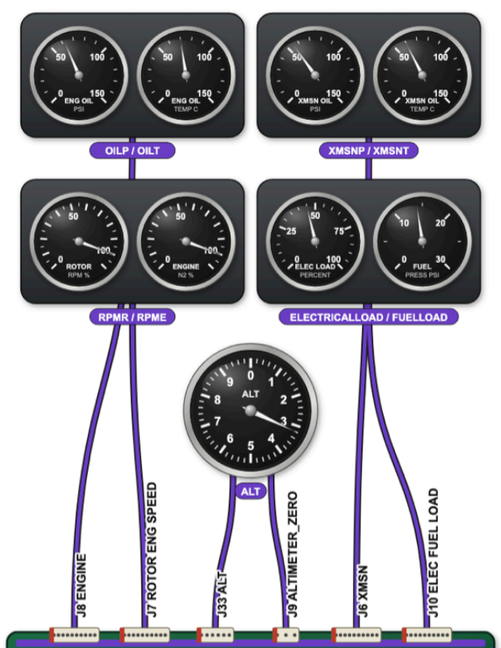
Jet Ranger cockpit: gauges, lamps and displays, and the controller board that drives each one

Instrument map, A3 landscape. Needles and lamps are drawn at illustrative readings. Cables end at the named connector on the board (references as printed on the PCBs).

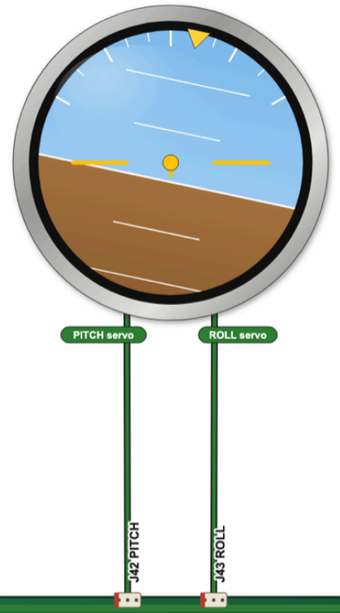
STEPPER GAUGES (X27 steppers)



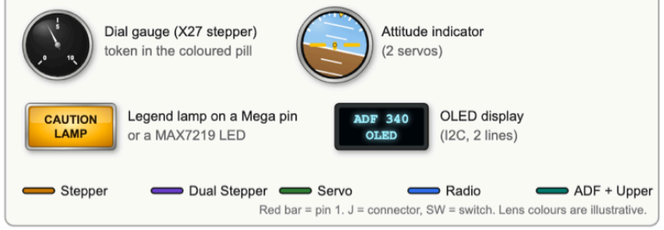
DUAL STEPPER GAUGES (dual instruments + altimeter)



ATTITUDE INDICATOR (pitch and bank servos)



KEY



HUEY WARNING PANEL (20 caution lamps on MAX7219 digits 4 - 7)

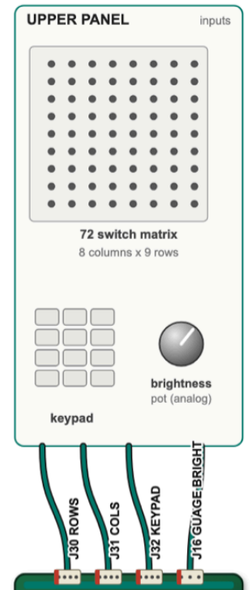
J20 - J23

ENG OIL PRESS	ENG ICING	ENG ICE JET	ENG CHIP DET	L FUEL BOOST	R FUEL BOOST	ENG FUEL PUMP	20 MIN FUEL	FUEL FILTER	GOV EMER
CL_A0	CL_B0	CL_C0	CL_D0	CL_E0	CL_F0	CL_G0	CL_H0	CL_I0	CL_J0
AUX FUEL LOW	XMSN OIL PRESS	XMSN OIL HOT	HYD PRESS	INLET AIR	INST INVERTER	DC GENERATOR	EXT POWER	CHIP DETECTOR	IFF
CL_A1	CL_B1	CL_C1	CL_D1	CL_E1	CL_F1	CL_G1	CL_H1	CL_I1	CL_J1

J22 HUEY ANODES · J23 HUEY CATHODES · J20 HUEY SWITCHES

ADF PANEL

Other switches and knobs and the USB HID board are inputs: not drawn.



STEPPER CONTROLLER

172.16.1.105

JET_RANGER_STEPPER_CONTROLLER

Arduino Mega 2560 + W5500 | Jet Ranger 13 Stepper v1.6

ETHERNET 172.16.1.0/24

DUAL STEPPER CONTROLLER

172.16.1.106

JET_RANGER_OLED_DUAL_STEPPER_CONTROLLER

Arduino Mega 2560 + W5500 | OLED Dual Stepper v1.5

ETHERNET 172.16.1.0/24

J55 BARO-OLED · J18 CLOCK-OLED · J17 ALTIMETER-OLED · J20 TEMPERATURE-OLED



DISPLAYS (4 OLEDs via a TCA9548A multiplexer) I2C D20 / D21

SERVO CONTROLLER

172.16.1.102

JET_RANGER_SERVO_CONTROLLER

Arduino Mega 2560 + W5500 | Front Panel Servo v1.4

ETHERNET 172.16.1.0/24

J4 LHS ANNUC · J5 RHS ANNUC

ANNUNCIATOR LAMPS (Mega pins A1 - A15)

ENGINE OUT (EOUT A1)	ROTOR LOW (RLOW A2)	XMSN OIL PRESS (TOPW A3)
XMSN OIL TEMP (TOTW A4)	BATTERY HOT (BHOT A5)	BATTERY TEMP (BTMP A6)
ENGINE CHIP (EC A7)	TAIL ROTOR CHIP (TRC A8)	XMSN CHIP (TC A9)
BAGGAGE DOOR (BD A10)	FUEL PUMP (FPMP A11)	AFT FUEL FILTER (FFLTR A12)
GEN FAIL (GENF A13)	LOW FUEL (LOWF A14)	SC FAIL (SCF A15)

RADIO CONTROLLER

172.16.1.101

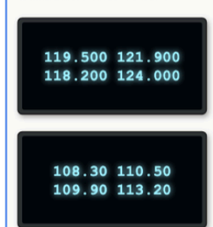
JET_RANGER_RADIO_CONTROLLER

Arduino Mega 2560 + W5500 | Radio Auto Pilot v1.4

ETHERNET 172.16.1.0/24

J8 COMM_NAV · J12 AUTOPILOT · J3 AUDIO

COM AND NAV PANEL



COM and NAV frequency OLEDs (8 encoders and 4 swap switches are inputs, not drawn)

AUTO PILOT PANEL 8 LEDs



AUDIO PANEL 11 LEDs



ADF + UPPER

172.16.1.107

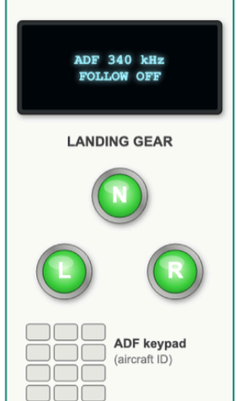
JET_RANGER_ADF_CONTROLLER

Arduino Mega 2560 + W5500 | Upper ADF v1.0

ETHERNET 172.16.1.0/24

J5 ADF

ADF PANEL



HOW THE VALUES ARRIVE

SIMULATOR
MSFS 2020 / 2024, Prepar3D, DCS World

↓

BRIDGE PROGRAM (on the PC)
FSUIPC, MSFS or DCS-BIOS bridge

↓

ETHERNET 172.16.1.0/24
UDP 13136: D,IAS:93.4,PITCH:4.5,...

↓

CONTROLLER BOARD
Mega 2560 + W5500, calibration tables

↓

CABLE TO THE CONNECTOR
stepper coils, servos, lamps, OLEDs

Each board takes real units and does its own calibration.

Sources: Project overview, Network and communication, PCB notes, Appendix B (COM ports), Appendix F (installing the controller PCBs) and Appendix H (Cockpit systems wiring) of the Jet Ranger master document. Diagrams are from `DocBuild/diagrams`.